Vehicle Longitudinal Dynamics Analysis

Bachelor's Degree in Automotive Engineering

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a.y 2023/2024

Abstract

The aim of this project is to analyze the performance of an internal combustion engine in order to accurately design the gear ratios and calculate the vehicle's acceleration, with the ultimate goal of estimating fuel consumption over a standardized driving cycle. To achieve this, an ICE performance map is utilized.

Contents

1 Introduction

This project consists of two main parts. In the first part, we analyze the power required for our vehicle to operate under different conditions. The second part focuses on the powertrain, aiming to determine the gear ratios that optimize vehicle performance across the entire operating range.

In the final stage, the results obtained are used to evaluate vehicle performance in real-world scenarios, such as acceleration times and fuel consumption. To do this, we calculate the net power, defined as the power delivered by the powertrain minus the power needed to overcome road grade, aerodynamic drag, and rolling resistance. If the net power is positive, the vehicle is accelerating; if it is negative, the vehicle is decelerating. When the net power is zero, the vehicle is traveling at a constant speed, meaning the acceleration is zero.

2 Power needed for motion

The first step is to use the provided data to calculate the resistance force for a range of different speeds and road grades using the following equation:

$$R = \left(mgcos\alpha - \frac{1}{2}\rho V^2 SC_z \right)(f_0 + kV^2) + \frac{1}{2}\rho V^2 SC_x + mgsin\alpha \tag{1}$$

which, since we neglect the aerodynamic lift, simplifies into:

$$R = A + BV^2 \tag{2}$$

where:

$$\begin{cases} A = mg(f_0\ cos\alpha + sin\alpha) \\ B = mgKcos\alpha + \frac{1}{2}\rho SC_x \end{cases} \tag{3}$$

The total power needed for motion can be now computed as follows:

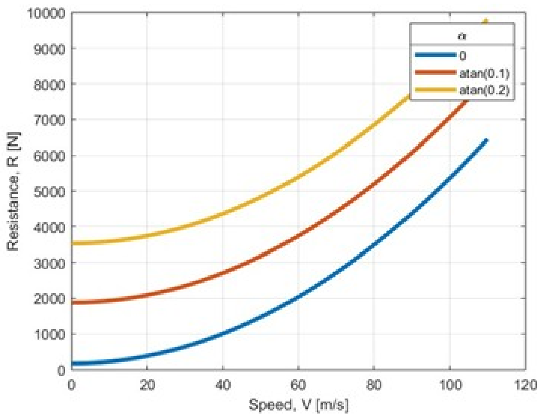
$$P_n = VR = AV + BV^3 \tag{4}$$

Moreover, the characteristic speed and power can also be evaluated with the following equations:

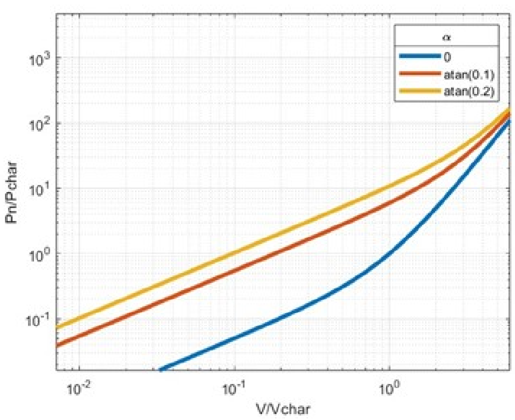
$$V_{char} = \sqrt{\frac{A}{B}} \tag{5}$$

$$P_{char} = 2A\sqrt{\frac{A}{B}} \tag{6}$$

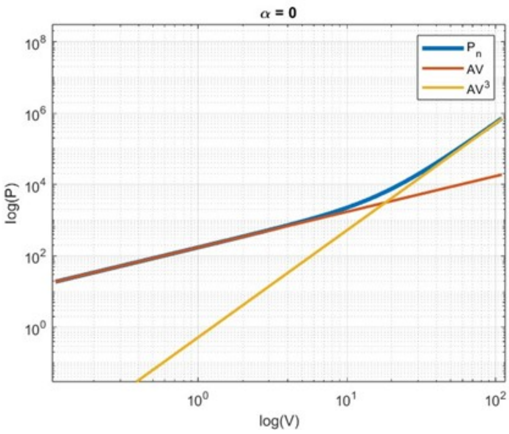
These steps are repeated for three different road grades, 0, 0.1, and 0.2, and the results are plotted on a logarithmic, normalized scale.



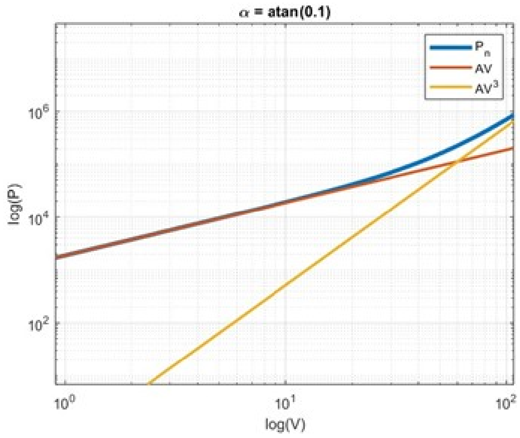
(a) Resistance force



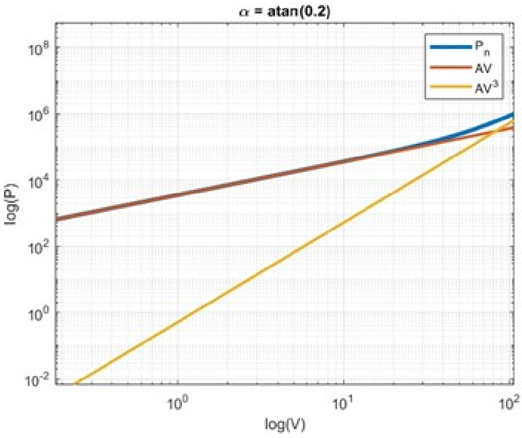
(b) Power demand



(a) α = 0



(b) α = atan(0.1)



(c) α = atan(0.2)

As expected, the resistance, and so the power needed for motion, increases with the road grade.

3 Maximum power available at the wheels

To approximate the vehicle's longitudinal behavior, it is also necessary to evaluate the power capability of the vehicle, which in this case is provided by an ICE engine. Using the available engine map, the power at the engine shaft can be calculated with the following formula:

$$P_e = \frac{mep\ V_{cil}\ \omega_e}{2\pi i} \tag{7}$$

where i is the number of cycles at each engine revolution. Since in this analysis a four strokes engine is considered, $i = 2$.

Moreover, the power delivered by an internal combustion engine at maximum throttle can be approximated by the Artamonov relationship:

$$\sum_{i=1}^{3} P_i\ \omega_e^i \tag{8}$$

This allows the computation of the torque and available power at the driving wheels, reported in Figures 3 and 4.

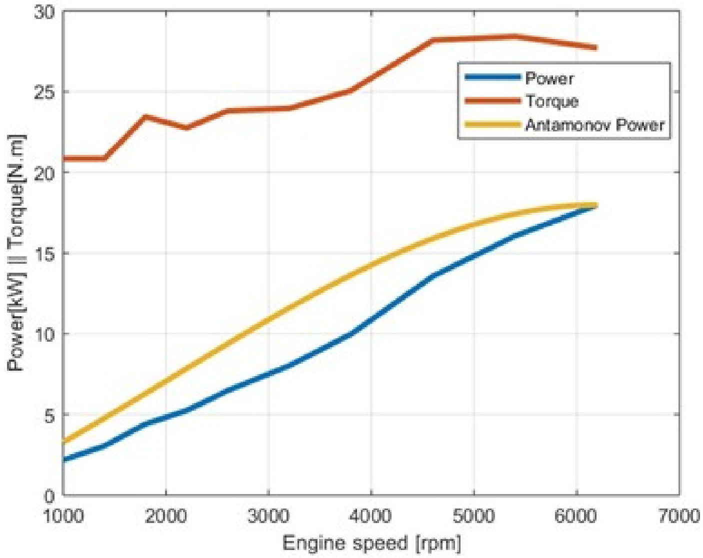


Figure 3: Engine torque and power

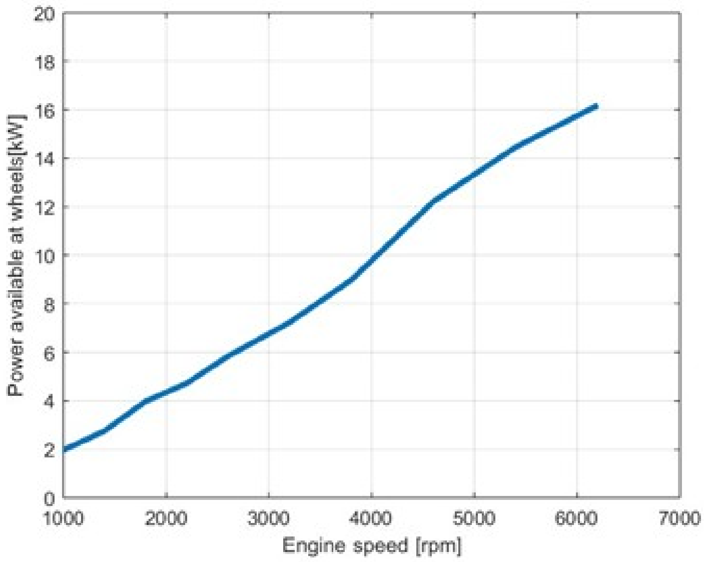


Figure 4: Power available at the wheels

It can be noticed that the power available at the wheels is slightly lower than the one provided by the engine due to the losses and non-unitary efficiency of the transmission.

4 Gradeability and transmission ratios

The next step is to design the gear ratios to maximize vehicle performance. First, the gear ratio that enables the vehicle to reach its maximum speed is calculated by equating the power required for motion to the maximum power available at the wheels. The top speed can be determined using Cardano's formula. Therefore the values of top gear ratio is evaluated as follows:

$$\tau_{g,top} = \frac{V_{max}}{\tau_f R_L \omega_{P,max}} \tag{9}$$

The bottom gear ration can be evaluated once the top one is known. It is of paramount importance that the bottom gear allows the vehicle to move in a certain speed range at a defined maximum grade. To this end, to enable the engine to start from rest, it is important that the minimum speed is included in this speed range. The assumption that the maximum grade is covered at reduced speed lead to the following expression that allows to compute the bottom gear:

$$\tau_{g,bottom} = \frac{M_e \eta_t}{\tau_f R_L mg(f_0\ cos\alpha_{max} + sin\alpha_{max})} \tag{10}$$

where $\alpha_{\max}$ is defined as the maximum grade:

$$i_{\max} = \tan(\alpha_{\max}) = 0.25$$

Once the top and bottom gear ratios have been determined, the intermediate ratios can be calculated using a geometric sequence defined as follows:

$$\tau_{g,i} = \tau_{g,i-1} \sqrt[N-1]{\frac{\tau_{g,top}}{\tau_{g,bottom}}} \tag{11}$$

The curves of wheel power as a function of speed can now be plotted for the different calculated gear ratios.

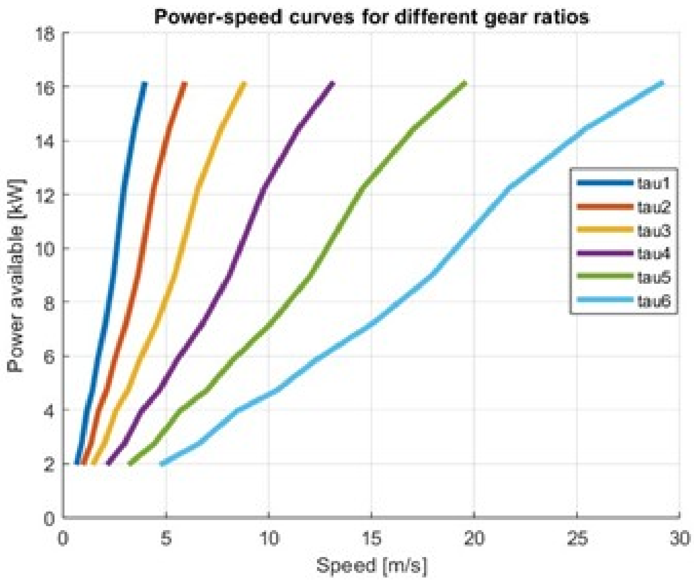


Figure 5: Power-speed curves for different gear ratios

As expected, increasing the gear ratio results in higher maximum and minimum achievable speeds. Consequently, the overall speed range expands with the gear ratio.

5 Maximum power that can be transferred from the tires

The maximum power at the tire-ground contact is first calculated using the following expression:

$$P_{max,tire} = V \sum_{1} \mu_{xi,p} F_{zi} \tag{12}$$

where the speed V is considered constant and F_{zi}, which refers to the vertical loads acting on each driving wheel, are calculated with the following equations:

$$F_{z1} = mg\frac{(b - \Delta x_2)cos\alpha - h_G sin\alpha - H_1 V^2}{l + \Delta x_1 - \Delta x_2} \tag{13}$$

$$F_{z2} = mg\frac{(a + \Delta x_1)cos\alpha + h_G sin\alpha - H_2 V^2}{l + \Delta x_1 - \Delta x_2} \tag{14}$$

where:

$$\begin{cases} H_1 = \frac{\rho S}{2mg} C_x h_G \\ H_2 = -H_1 \\ \Delta x_i = R_{Li}(f_0 + KV^2) \end{cases} \tag{15}$$

Moreover, since the longitudinal force coefficient μ_{xp} varies with the longitudinal speed, it can be expressed as follows:

$$\mu_{xp} = c_1 - c_2 V \tag{16}$$

where c_1 and c_2 are known coefficients.

The following graph shows the maximum achievable speeds for dry and wet conditions, which correspond to the intersection points between the required power and tire power curves. The results indicate $V_{max,dry} = 98$ m/s and $V_{max,wet} = 69$ m/s. This is consistent with theoretical expectations, as under wet conditions the power available at the tires is lower than in dry conditions, as shown in the chart.

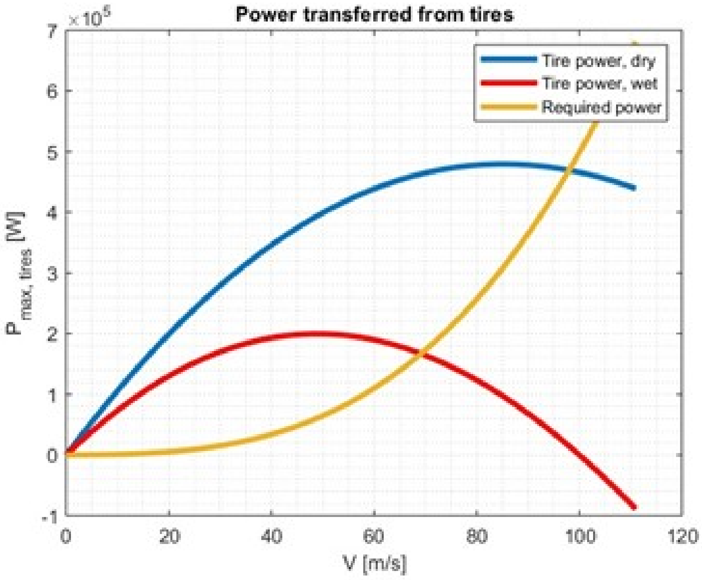


Figure 6: Power transferred from tires

6 Acceleration performances

Acceleration can be calculated by evaluating the difference between the available and required power, which corresponds to the derivative of the kinetic energy, as shown by the following equations:

$$P_a - P_n = \frac{dT}{dt} \tag{17}$$

$$T = \frac{1}{2} m_e V^2 \tag{18}$$

where the term m_e represents the equivalent mass, accounting for the inertia of the wheels, powertrain, and engine. It is calculated using the following formula:

$$m_e = m + \frac{J_w}{R_L^2} + \frac{J_t}{R_L^2 \tau_f^2} + \frac{J_e}{R_L^2 \tau_f^2 \tau_g^2} \tag{19}$$

Therefore, the maximum acceleration can be written as:

$$a_{max} = \frac{P_{a,max} - P_n}{V m_e} \tag{20}$$

The acceleration–speed characteristics can be plotted for each gear ratio. This allows the determination of the optimal shift points to maintain the highest available acceleration. Consequently, the time required to reach a reference speed can be calculated by integrating the inverse of the acceleration curve with respect to speed.

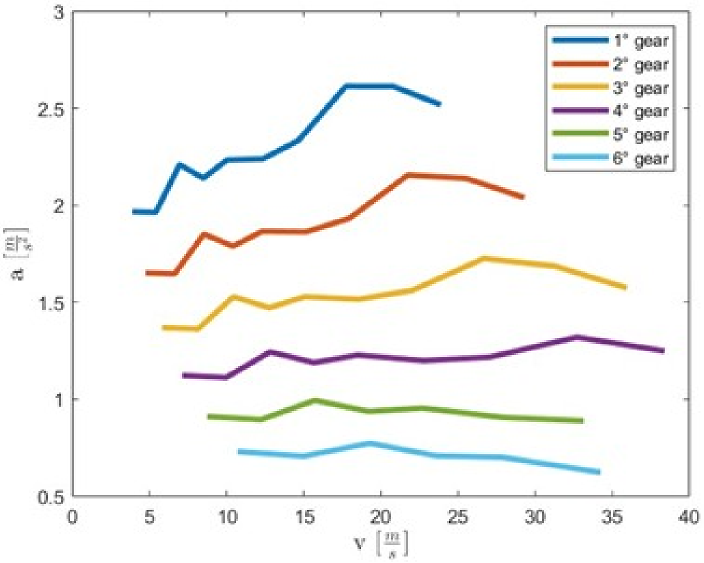


Figure 7: Acceleration curves for different gear ratios

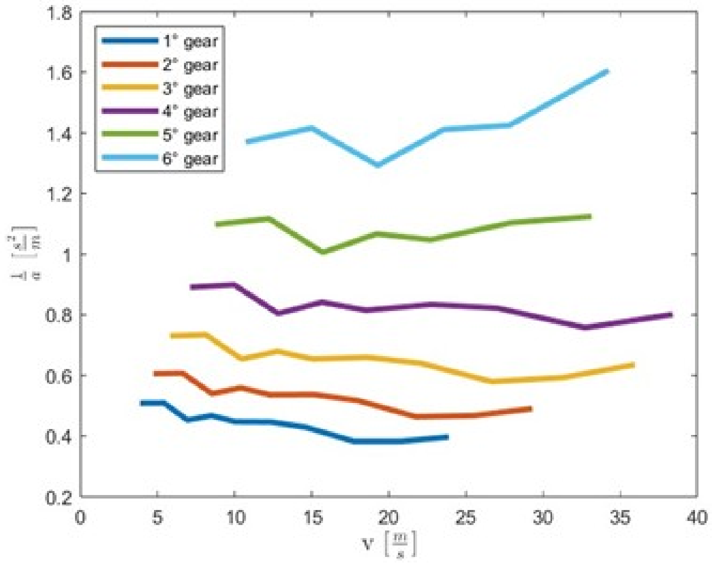


Figure 8: Function $1/a$ for different gear ratios

Analysis of the two graphs shows that the acceleration curves differ slightly from the theoretical expectations. This discrepancy is likely due to inaccuracies in the engine data used. In particular, the torque–speed characteristic does not exhibit the typical parabolic shape of an ICE engine, indeed it is almost constant. Nevertheless, the resulting acceleration curves are consistent with the torque data.

7 Computation of fuel consumption

To calculate fuel consumption, the total power must first be determined, defined as the sum of the power required for motion and the power needed to accelerate the vehicle:

$$P_{tot} = P_n + P_{acc} = RV + m_e\ a\ V_m \tag{21}$$

The engine power can then be easily determined, allowing the calculation of the mean effective pressure:

$$mep = \frac{P_e 2\pi i}{V_{cil} V_m} \cdot \tau_g \tau_f R_L \tag{22}$$

Once the mean effective pressure is determined, the specific fuel consumption can be calculated using linear interpolation based on the available data. The fuel consumption is then evaluated for each segment of the NEDC cycle, and the contributions are summed to obtain the total fuel consumption for the vehicle over the analyzed driving cycle. The result is Q = 8.78 l/100km.

8 Conclusions

In conclusion, aside from the torque–speed characteristic, the data and plotted charts are consistent. Furthermore, the calculated fuel consumption represents realistic values for the engine and vehicle considered in this study.